

NEW UZBEKISTAN UNIVERSITY

Week 3

Data Cleaning and Quality

Decisions that change what
data can tell us

Mohammad Sabokrou
Data Science Fundamentals



Week 3 learning objectives

CLO1: explain preprocessing. CLO2: clean data appropriately.

Quality dimensions	Explain six quality dimensions and choose checks for a stated question.
Cleaning decisions	Justify format fixes and duplicate rules. Distinguish unusual values from errors.
Missingness	Explain MCAR, MAR, and MNAR using examples and state the assumptions.
Audit trail	Record each decision, count its effect, and check the cleaned data with assertions.

The connection to Week 2

A summary depends on which values enter the calculation

Last week

We described centre, spread, and shape.

We reported the unit and the number of valid observations.

This week

We inspect the values behind those summaries.

We ask whether cleaning changes the population or the conclusion.

A correct formula can still summarize the wrong data.

A registry export arrives

Illustrative records. One row should describe one student in the current registry.

ID	Programme	Age	Score	Updated
A01	Biology	19	72	2026-09-01
A02	biology	20	67%	2026-09-01
A02	Biology	20	69	2026-09-03
A03	Economics	220	blank	2026-09-01
A04	Economics	47	88	2026-09-01

Discussion: What needs investigation before you calculate an average?

One repeated record changes an average

Illustrative scores. The second A02 row repeats the same measurement.

Record	Score
A01	40
A02	80
A02, repeated copy	80

Mean of rows = 66.7

Mean of students = 60.0

The repeated record gives one student twice the weight.

Quality is fitness for a purpose

A cell holds one value. This student table contains one missing phone-number cell.

Student ID	Programme	Phone
101	AI	555-0101
102	AI	(missing)
103	Robotics	555-0103

Count students by programme
AI: 2. Robotics: 1. The blank phone does not affect these counts.

Make urgent phone calls
Student 102 cannot be called using this table. The blank matters here.

Data quality depends on whether the data support the intended task.

Six dimensions of data quality

Required records, valid values, and agreement between fields

Completeness

Check missing cells and absent records. Example: an export contains 95 of 100 expected students.

Validity

Do values follow a documented type, category, or range? Example: score from 0 to 100.

Consistency

Do related values agree? Example: an end date should not precede its start date.

Six dimensions of data quality

Second: identity, reality, and time

Uniqueness

A key identifies a row. A current student registry expects one row per student ID.

Accuracy

Does the recorded value match the true value? Verification may require the original record.

Timeliness

Is the information recent enough for this decision? Check the event date and snapshot date.

A valid value can still be wrong

Valid means it follows a rule. Accurate means it matches the correct value for that student.

Stored score	0-100 range check	Verified marked exam: 76
760	Fails	Outside the permitted score range.
67	Passes	Wrong: someone reversed the digits of 76.
76	Passes	Matches the verified exam record.

Why? A range check accepts both 67 and 76. It cannot tell which score the student earned.

An inventory records the starting condition

An inventory is a short summary before cleaning. It reveals problems and lets us explain later changes.

Check	Example before cleaning	Why record it?
Rows and IDs	12 rows, 10 distinct student IDs	Repeated IDs need investigation.
Missing values	2 of 12 phone cells are missing	Compare missingness before and after.
Types and ranges	Age includes 220 and text "21"	Check the age error and the stored type.

After cleaning: 12 rows become 10 only if we can explain the two removed rows.

The data dictionary defines the checks

Example dictionary for an illustrative registry

Field	Meaning	Documented rule
student_id	Identifier	Text. Required and unique in current registry.
entry_score	Admission result	Numeric points from 0 to 100.
programme	Degree programme	One of the approved programme names.
updated_at	Record revision time	Timestamp with documented timezone.

Different labels can describe one category

Software counts different text labels as separate groups. This can make Biology look like three programmes.

Raw programme label	Students	Meaning confirmed by dictionary
" biology "	4	Biology. Remove outside spaces.
"Biology"	3	Biology. Keep the approved label.
"BIO"	2	Biology. Apply the approved abbreviation map.
"Bioengineering"	1	A different programme. Keep it separate.

After standardising: Biology = 4 + 3 + 2 = 9 students. Bioengineering = 1.

Stored types and units affect calculations

Type: text or number. Unit: what the number measures. Representation: how we write it. We need numeric values to calculate a mean, and matching units to compare them.

Raw text	Meaning	Correct stored value
"67%"	Percentage	Number 0.67 as a proportion
"0.67"	Proportion from 0 to 1	Number 0.67, equivalent to 67%
"00124"	Student identifier	Text "00124" keeps leading zeros
"1,200"	Separator convention unclear	Confirm whether it means 1200 or 1.2

The data dictionary explains meanings and units. Conversion follows that definition.

A date parser needs the source format

A parser is software that reads text using a stated format. It turns date text into a date so we can sort events or calculate durations. What does “03/04/2026” mean?

Format supplied to the parser	Date it reads	Clear output
Day / month / year	3 April 2026	2026-04-03
Month / day / year	4 March 2026	2026-03-04

A wrong choice moves the event by 30 days. Both results are possible dates.

Check the source’s date convention first. Keep unresolved dates flagged until confirmed.

Numeric conversion and new missing values

A numeric parser turns text such as “72” into the number 72 for calculations. In this example, failed conversions become missing-value markers.

Raw score text	Numeric result	What happened?
“72”	72	Conversion succeeds.
(blank)	missing	The raw file already lacked a value.
“seventy”	missing	New missing value: this parser cannot read words.

Missing cells increase from 1 to 2. Keep the raw text and investigate the failed conversion.

Three meanings of repeated records

The unit of observation means what one row represents, such as one student or one weekly result.

Case	Two example records	Appropriate response
Exact copy	ID 101, score 70 ID 101, score 70	For a current registry: confirm a copied row, then keep one.
Conflicting ID	ID 102, score 60 ID 102, score 80	Verify the correct score or version. Keep the conflict flagged meanwhile.
Valid repetition	ID 103, week 1, score 70 ID 103, week 2, score 80	For weekly results: keep both. The key includes ID and week.

A repeated student ID is a problem only when the intended row definition requires it to be unique.

A conflicting ID needs a rule

Illustrative revisions from a system that explicitly tracks corrections

ID	Score	updated_at	Status
A02	67	2026-09-01 09:00	Original
A02	69	2026-09-03 11:00	Verified correction

Here, the source confirms that the verified correction supersedes the original.

Keep the original in the audit history.

When “keep the latest” is unsafe

A timestamp only helps when its meaning is clear

- Registration date** The student registered later. That does not prove the score is a correction.
- Conflicting sources** A recent spreadsheet may be less authoritative than the signed record.
- Unresolved conflict** Retain both versions for review and flag the field or record as unresolved.

The cleaning rule should name the evidence that makes one record authoritative.

Discussion: which repeated rows should stay?

These three small tables have different row meanings. For each pair, choose a key and justify an action.

Table	Row	ID / invoice	Other recorded fields
Enrolments	A	Student 101	Course DS
Enrolments	B	Student 101	Course CV
Purchases	C	Invoice 501	Same purchase, total 100 USD
Purchases	D	Invoice 501	Same purchase, total 100 USD
Current registry	E	Student 102	Birth date 2006-02-10
Current registry	F	Student 102	Birth date 2006-10-02

Which pair stays, which confirmed copy can go, and which needs source evidence? Why?

Reasoning about repeated rows

A useful answer includes the unit and the evidence

Two courses

Keep both registrations. Student ID alone is not the registration key.

Duplicate load

Keep one purchase after verifying that both rows represent the same invoice event.

Birth-date conflict

Check an authoritative source. An arbitrary first or last row has no evidential advantage.

Question 1: is the reported mean justified?

Five minutes. One current row per student. BIO and “ biology ” mean Biology.
The source has not confirmed which score for 00102 is correct. Scores use 0–100 points.

Row	Student ID	Programme	Score	Recorded date
A	00101	Biology	70	2026-09-01
B	00101	BIO	70	2026-09-01
C	00102	biology	60	2026-09-01
D	00102	Biology	80	2026-09-03
E	00103	Bioengineering	90	2026-09-01

B is a confirmed copy of A. An analyst keeps the latest row per ID and reports Biology mean = 75.
1. Is 75 justified? Explain. 2. Which changes are safe, and which need evidence?
3. Using the two candidate scores for 00102, calculate both possible Biology means.

Email during class to m.sabokrou@newuu.uz. Subject: STUDENTID_W3_Q1
Body: class group, reasoning in your own words, and calculations. About 80-120 words.

Missing values

A blank can change both the denominator and the conclusion

Who did not appear in this number?

The reason for missingness matters as much as the amount.

Zero, missing and not applicable differ

Example: “How many hours did you use the campus laboratory this week?”
The question applies only to students enrolled in the campus lab module.

Student	Recorded hours	Meaning
101: enrolled	0	Tracked for the week. No lab use occurred.
102: enrolled	(missing)	Tracking failed. Actual hours could be 0, 2 or another value.
103: not enrolled	Not applicable	Outside this question’s target group.

Changing student 102’s blank to 0 claims “no use”, although we do not know the hours.

Completeness needs a denominator

Course-satisfaction survey: 1 = very dissatisfied, 5 = very satisfied. We invite 20 students.

Quantity	Value
Students invited	20
Students who answered	12
Missing answers	8
Completeness of answers	$12 \div 20 = 60\%$

Completeness tells us how much is missing.

A mechanism describes why answers are missing and who is less likely to reply.

MCAR: loss unrelated to students or answers

MCAR means missing completely at random. Example: random technical loss of survey forms.

Assumed process	What it means
20 ratings collected, 8 randomly lost	12 remain. Completeness = $12 \div 20 = 60\%$.
Every form has the same loss chance	Loss ignores class group and satisfaction rating.
We analyse the 12 remaining ratings	No systematic selection toward higher or lower ratings.

Completeness: 60% of answers remain. MCAR: the loss process is random. Fewer answers mean greater uncertainty in the estimate.

MAR: reply chance depends on known class

MAR = missing at random given observed information. Class is known for everyone. Missing value: course satisfaction. The table gives assumed reply chances.

Known class	Satisfied: reply chance	Dissatisfied: reply chance
Day	90%	90%
Evening	40%	40%

Suppose evening students have less time to reply. Within either class, satisfied and dissatisfied students have the same chance of answering.

The missing rating does not change reply chance after we account for class.

MNAR: satisfaction still affects replies

MNAR means missing not at random. This table assumes a different evening reply process. We still know every student's class, but nonresponders' ratings remain unknown.

Known class	Satisfied: reply chance	Dissatisfied: reply chance
Day	90%	90%
Evening	40%	10%

Suppose dissatisfied evening students avoid the survey. Even within evening class, their chance of replying is lower. Replies can make satisfaction look higher.

MAR on the previous slide: $40\% = 40\%$. MNAR here: $40\% \neq 10\%$ within evening class.

MCAR, MAR and MNAR describe missingness

A mechanism describes the process that makes a value missing. These names classify assumptions.

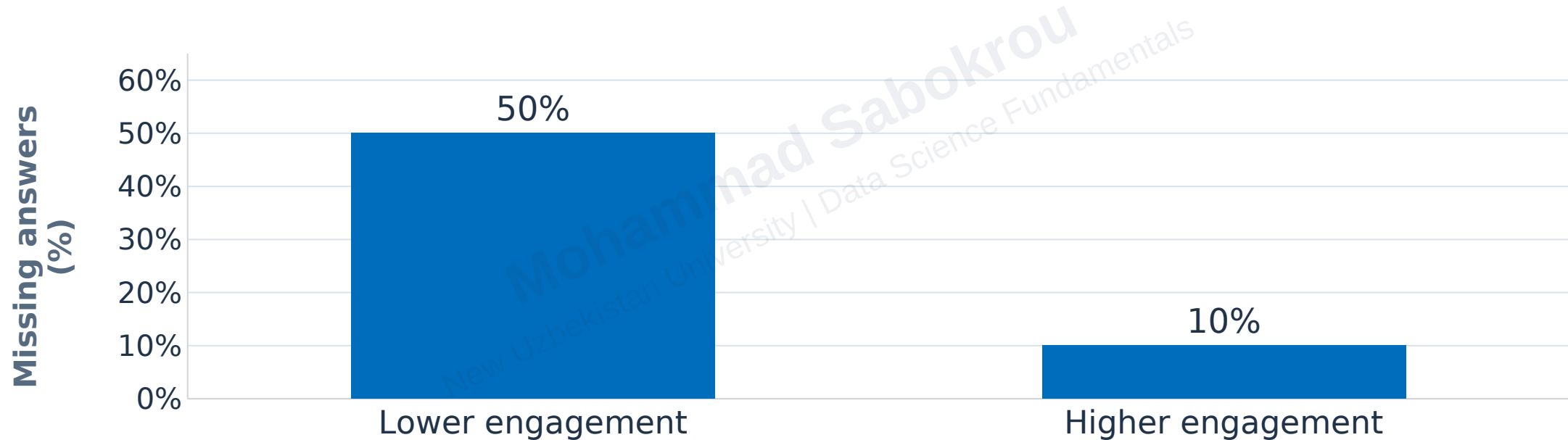
Same survey throughout: known class, potentially missing course-satisfaction rating.

Assumption	Example process	What changes reply chance?
MCAR	Random loss of forms	Neither class nor satisfaction.
MAR	Evening students have less time	Class matters. Within class, satisfaction does not.
MNAR	Dissatisfied evening students avoid replying	Satisfaction matters even within the same class.

These are assumptions about missingness, not measured quality scores. Observed answers alone generally cannot tell MAR from MNAR.

Different response rates need an explanation

Course-satisfaction survey: 20 students per group. Here, engagement means attendance. Lower group: 5 of 10 classes attended. Higher group: 9 of 10 classes attended.



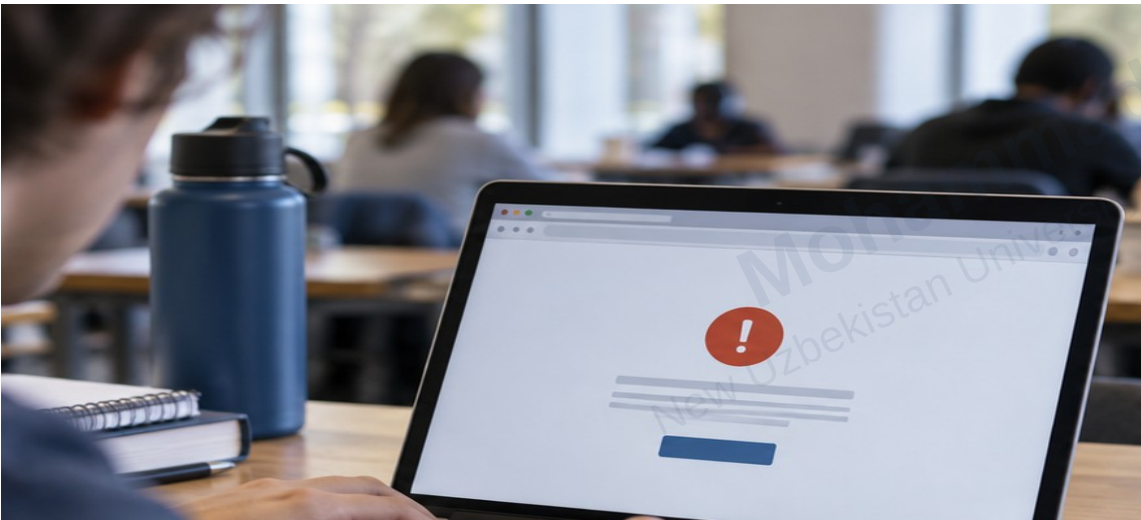
Missing ratings: lower group $10/20 = 50\%$, higher group $2/20 = 10\%$.

Possible causes: missed reminders or dissatisfaction. The rates do not prove MAR.

The same missing rate can hide different bias

Each illustrative programme invites 20 students and has 12 satisfaction ratings.
Missing: $8/20 = 40\%$ in both programmes.

A: random technical loss



Eight randomly chosen forms are lost.
No systematic preference for satisfied students.

B: dissatisfied students skip



The eight most dissatisfied students skip.
The replies make the programme look happier.

Deleting incomplete rows changes who remains

Illustrative class. A required survey field has different response rates.

Group	Original students	Complete rows	Retained
Day group	80	72	90%
Evening group	20	8	40%
Total	100	80	80%

Evening students fall from 20% of the class to 10% of the complete cases.

Mean imputation reduces the visible spread

Imputation fills missing values. Here, three blanks receive the observed mean of 3.

Treatment	Displayed values	Mean	Sample SD
Observed values only	2, 4	3	1.41
Fill blanks with 3	2, 4, 3, 3, 3	3	0.71

The mean is unchanged. The apparent variation is smaller.

Five filled cells still contain only two observed answers.

Sensitivity analysis makes assumptions visible

20 invited students. 12 responses have mean 4.0 on a 1-5 scale.

$$\text{Observed total} = 12 \times 4 = 48$$

Let m be the average answer of the 8 missing students.

$$\text{Mean for all 20} = (48 + 8 \times m) \div 20$$

We vary m to see whether the conclusion depends on answers we do not know.

Different assumptions, different conclusions

The target is an overall mean of at least 3.5

Assumed missing mean m	Overall mean	Meets 3.5?
1.0	2.8	No
3.0	3.6	Yes
4.0	4.0	Yes
5.0	4.4	Yes

With scores limited to 1-5, the overall mean can range from 2.8 to 4.4.

Recover the missing value from its source

A registry export has no score for student 201. The verified original exam record gives 78.

Record	Raw export	After source verification
Student 201 score	missing	78 points

We recover an existing measurement. Match the student and exam before copying the value.

Keep the missing value and report the limit

Scores are 70, 80 and missing for three students. We want to describe what we actually observed.

Quantity	Result
Mean of observed scores	$(70 + 80) \div 2 = 75$
Coverage	2 observed scores out of 3 students

We can report “observed mean = 75, n = 2 of 3”. We cannot claim the all-student mean is 75.

Exclude rows for a stated analysis

The same scores are 70, 80 and missing. An analyst keeps only rows with a recorded score.

Student	Score	Included in this analysis?
201	70	Yes
202	80	Yes
203	missing	No

The result describes two included students. Excluding a row can change who the analysis represents.

Impute an assumed replacement value

Imputation fills a blank with an estimated or assumed value. Here, the analyst fills it with the observed mean.

Student	Observed score	Stored score after filling
201	70	70
202	80	80
203	missing	75, imputed

The mean stays 75, but student 203's score is still unknown. Mark the filled value as imputed.

Summary: four responses to missing data

Lab 3: diagnose missingness and keep missing values unfilled.

Option	When it can help	What to report
Recover the value	A reliable source or follow-up exists.	Source and recovery process.
Leave it missing	Analysis can use other observed fields.	Valid n and missingness pattern.
Exclude observations	The target and missingness assumptions justify it.	Retention and group differences.
Impute values	A justified model and uncertainty method exist.	Assumptions and sensitivity.

Unusual values need an explanation

Illustrative ages in an undergraduate registry

Age	Interpretation	Initial response
220	Impossible human age.	Check source. Mark invalid if unresolved.
47	Unusual but plausible student.	Keep and verify if necessary.
21	Plausible, but may still be mistyped.	Accuracy needs source evidence.

Distance from the average does not determine whether a record is wrong.

The IQR rule flags values for review

Q1 (25th percentile) = 19. Q3 (75th percentile) = 22. Ages are in years.

$$\text{IQR} = \text{Q3} - \text{Q1} = 3 \text{ years}$$

Lower fence

$$\begin{aligned} \text{Q1} - 1.5 \times \text{IQR} \\ 19 - 1.5 \times 3 = 14.5 \end{aligned}$$

Upper fence

$$\begin{aligned} \text{Q3} + 1.5 \times \text{IQR} \\ 22 + 1.5 \times 3 = 26.5 \end{aligned}$$

Both ages 47 and 220 exceed the upper fence. Their explanations are very different.

A threshold can remove a subpopulation

Illustrative cohort. All 5 students above age 35 study part-time.

Population	Full-time	Part-time	Total
Original cohort	80	20	100
After deleting age > 35	80	15	95

Part-time share changes from 20.0% to 15.8%.

The cleaned table now represents a different mix of students.

Influence can be reported without deletion

Withdrawal means leaving the course. The goal is to describe all 100 enrolled students. Five students are older than 35. Their ages are valid, and four of them withdrew.

Students counted	Withdrawals	Students	Withdrawal rate
All students	10	100	$10 \div 100 = 10.0\%$
Older than 35	4	5	$4 \div 5 = 80.0\%$
After excluding the older group	$10 - 4 = 6$	$100 - 5 = 95$	$6 \div 95 = 6.3\%$

Deleting the five older students removes four withdrawals: 10% becomes 6.3%. This changes the population. Keep the all-student result and explain the comparison.

A defensible response to an outlier

The evidence determines the action

Verified entry error	Correct it from the source. Preserve the raw value and record the reason.
Invalid, unresolved value	Flag the value and retain usable fields where appropriate. Do not guess a replacement.
Valid, influential value	Keep it. Use suitable summaries and report sensitivity when the result changes.

Question 2: does this survey meet the target?

Five minutes. Satisfaction uses 1–5. The course target is an all-student mean of at least 3.5. An analyst pools the replies and claims: “Mean = 3.8, so the course meets the target.”

Known class	All students	Replies	Missing	Reply mean
Day	80	72	8	4
Evening	20	8	12	2

1. Assume replies represent each class. Estimate the mean using the full class sizes.
2. Do different reply rates prove MAR? Explain what must hold within each class.
3. Now assume the 12 missing evening ratings average 1, and all day students average 4. Recalculate the all-student mean. What happens to the target claim?

Email during class to m.sabokrou@newuu.uz. Subject: STUDENTID_W3_Q2

Body: class group, reasoning in your own words, and calculations. About 80-120 words.

Examples behind the cleaning log

Issue = what needs attention. Affected = how many rows or values.

Action = what you changed or kept. Reason = the evidence for that decision.

Before	Action and result	Reason
Four rows: " biology "	Trim spaces and use "Biology"	Dictionary confirms the same programme.
One confirmed copied row	Keep one copy, remove the extra row	Both rows describe the same record.
One student: age 220	Keep student, flag age as missing	Invalid age, correct age unknown.
One student: age 47	Keep the row and age	Unusual but plausible for this population.

Log the units: 4 rows edited; 1 copy removed; 1 value flagged; 1 record retained.

Next: the summary cleaning log.

The cleaning log explains every decision

An illustrative audit record

Issue	Affected	Action	Reason
Whitespace in programme	4 rows	Trim spaces	Same approved category.
Verified duplicate load	1 extra row	Retain one	Same record and event.
Age 220	1 value	Flag as missing	Invalid. No verified correction.
Age 47	1 record	Keep	Plausible member of target cohort.

Assertions and accuracy checks

Assertions stop on a failed rule. Accuracy checks compare with a trusted source.

Rule or source check	Example check result	Next action
Current ID must be unique	2 IDs repeat	Inspect conflicts before analysis.
Score must lie from 0 to 100	1 score equals 760	Review and resolve or flag.
Programme must be approved	1 label is unknown	Check dictionary or source.
Compare with the original record	Not yet checked	Verify or report the limitation.

Row accounting separates different changes

Illustrative registry export with 12 rows

Step	Rows	What changed
Raw export	12	Starting point.
Remove 2 confirmed exact copies	10	Two extra copies removed.
Resolve 1 verified earlier version	9	One current row per student.
Flag 1 invalid age as missing	9	One age changes. Student stays.

A cleaning log should explain both row removals and value changes.

Case discussion: a small registry

One row per current student. Scores use a documented 0–100 scale.

ID	Programme	Age	Score	Record status
B01	Biology	19	70	Current
B01	Biology	19	70	Confirmed extra copy
B02	biology	220	80%	Current
B03	Economics	47	blank	Current

Choose actions. Then report the mean score and its limitation.

Case discussion: a defensible solution

Three students remain after removing the confirmed extra copy

ID	Programme	Age	Score
B01	Biology	19	70
B02	Biology	missing, flagged	80
B03	Economics	47	missing

Observed score mean = $(70 + 80) \div 2 = 75$

This is the mean for 2 observed scores among 3 students.

A result needs an interpretation and a limit

Example reporting paragraph for the small registry

“After removing one confirmed duplicate, the registry contains three students. Two have recorded entry scores, with a mean of 75 out of 100. One score remains missing. We corrected documented formatting and flagged an invalid age. We cannot conclude that the mean score of all three students is 75.”

The paragraph identifies the population, denominator, action, and uncertainty.

What should survive every cleaning step?

A short explanation is part of the analysis

Meaning

The variable, unit, population, and key still have clear definitions.

Evidence

Each correction or exclusion has a reason that another analyst can inspect.

Uncertainty

Missingness, unresolved conflicts, and the effect of assumptions remain visible.

A defensible cleaning decision can be explained in an ordinary sentence.

Questions to take away

Explain your reasoning before looking back at the worked examples

- 1** A value is four standard deviations from the mean. What evidence would justify removal?
- 2** Two groups have the same response rate. What could still bias their comparison?
- 3** All checks pass. Which kind of quality problem may still remain?

Lab 3 and the next lecture

Chapter 4: Data Quality and Cleaning

Lab 3: registry cleaning

Submit the notebook, cleaning log, and cleaned CSV.

Inventory issues, justify each fix, and run assertions.

Keep missing values unfilled.

Reading: Chapter 4. Record the evidence and limitation for each cleaning decision.

Week 4

Transformation, features, and leakage.

How representation and the order of preprocessing affect an analysis.

References

Course material and technical documentation

Course plan and Fundamentals of Data Science, Vol. 1, Chapter 4

Seaman et al. (2013): What Is Meant by “Missing at Random”?

NIST/SEMATECH handbook: Outliers (§7.1.6), Box Plot (§1.3.3.7)

pandas: `to_numeric` and `DataFrame.drop_duplicates`

All short classroom datasets in this lecture are illustrative.